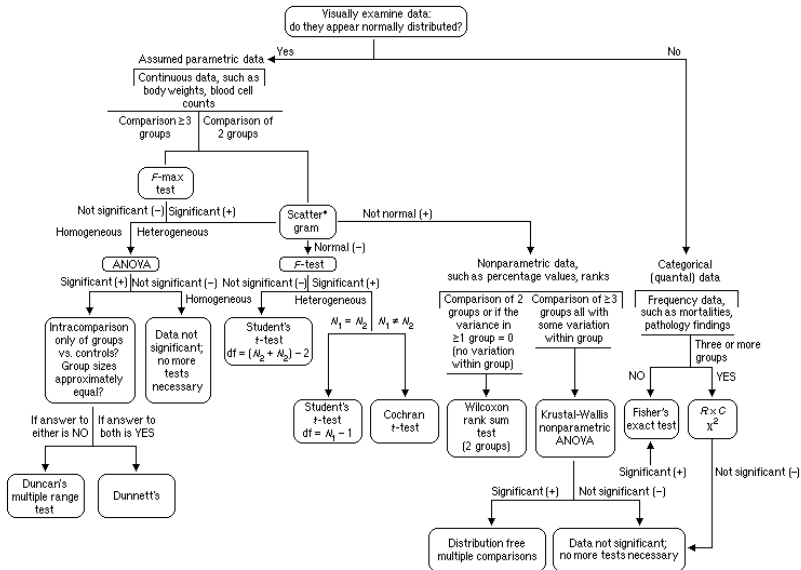


GLM as a unified framework for data analysis

Francisco Rodríguez-Sánchez

<https://frodriguezsanchez.net>

Statistics as a maze



So many questions

- **Why** should we really use analysis Y over Z?

So many questions

- **Why** should we really use analysis Y over Z?
- What if my data are **not Normal**?

So many questions

- **Why** should we really use analysis Y over Z?
- What if my data are **not Normal**?
- What if they are **not independent**?

So many questions

- **Why** should we really use analysis Y over Z?
- What if my data are **not Normal**?
- What if they are **not independent**?
- Why am I getting **different p-values** with different tests?

So many questions

- **Why** should we really use analysis Y over Z?
- What if my data are **not Normal**?
- What if they are **not independent**?
- Why am I getting **different p-values** with different tests?
- What even is a **p-value**?

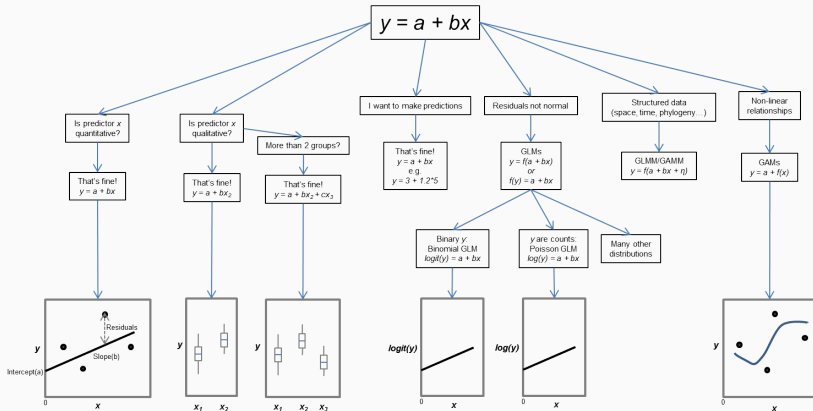
So many questions

- **Why** should we really use analysis Y over Z?
- What if my data are **not Normal**?
- What if they are **not independent**?
- Why am I getting **different p-values** with different tests?
- What even is a **p-value**?
- How can I take **different factors** into account?

So many questions

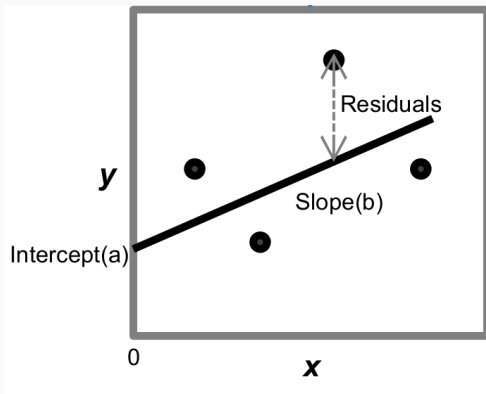
- **Why** should we really use analysis Y over Z?
- What if my data are **not Normal**?
- What if they are **not independent**?
- Why am I getting **different p-values** with different tests?
- What even is a **p-value**?
- How can I take **different factors** into account?
- Can I make **predictions**?

A unified framework



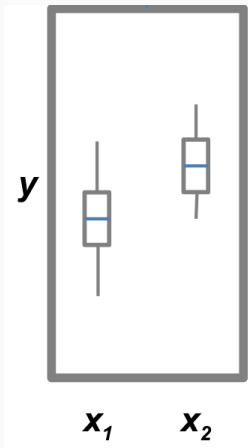
Linear regression

$$y = a + bx$$



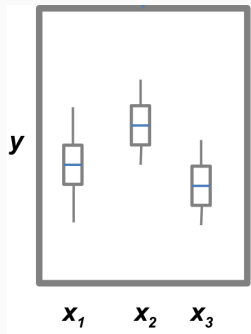
Is predictor X qualitative?

$$y = a + bx_2$$



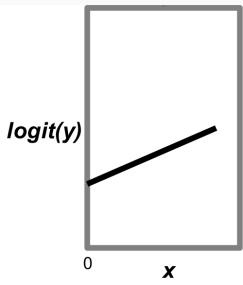
More than 2 groups?

$$y = a + bx_2 + cx_3$$



My data (residuals) are not Normal

$$y = f(a + bx)$$

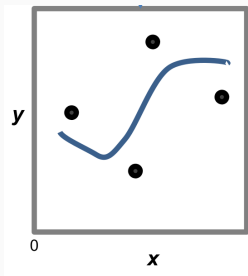


My data are structured (space, time, phylogeny)

$$y = f(a + bx + \eta)$$

Relationships are not linear

$$y = a + f(x)$$



t-tests

ANOVA

regression

...

are special cases of GLM

Common statistical tests are linear models

Last updated: 02 April, 2019

See worked examples and more details at the accompanying notebook: <https://lindeloev.github.io/tests-as-linear>

	Common name	Built-in function in R	Equivalent linear model in R	Exact?	The linear model in words	Icon
Simple regression: $\text{lm}(y \sim 1 + x)$	y is independent of x P: One-sample t-test N: Wilcoxon signed-rank	t.test(y) wilcox.test(y)	$\text{lm}(y \sim 1)$ $\text{lm}(\text{signed_rank}(y) \sim 1)$	✓ for $N \geq 14$	One number (intercept, i.e., the mean) predicts y. - (Same, but it predicts the signed rank of y.)	
	P: Paired-sample t-test N: Wilcoxon matched pairs	t.test(y1, y2, paired=TRUE) wilcox.test(y1, y2, paired=TRUE)	$\text{lm}(y_2 - y_1 \sim 1)$ $\text{lm}(\text{signed_rank}(y_2 - y_1) \sim 1)$	✓ for $N \geq 14$	One intercept predicts the pairwise $y_2 - y_1$ differences. - (Same, but it predicts the signed rank of $y_2 - y_1$.)	
	y ~ continuous x P: Pearson correlation N: Spearman correlation	cor.test(x, y, method='Pearson') cor.test(x, y, method='Spearman')	$\text{lm}(y \sim 1 + x)$ $\text{lm}(\text{rank}(y) \sim 1 + \text{rank}(x))$	✓ for $N \geq 10$	One intercept plus x multiplied by a number (slope) predicts y. - (Same, but with ranked x and y)	
	y ~ discrete x P: Two-sample t-test N: Welch's t-test N: Mann-Whitney U	t.test(y1, y2, var.equal=TRUE) t.test(y1, y2, var.equal=FALSE) wilcox.test(y1, y2)	$\text{lm}(y \sim 1 + G_1)^a$ $\text{glm}(y \sim 1 + G_1, \text{weights} = \dots)^a$ $\text{lm}(\text{signed_rank}(y) \sim 1 + G_1)^a$	✓ ✓ for $N \geq 11$	An intercept for group 1 (plus a difference if group 2) predicts y. - (Same, but with one variance per group instead of one common.) - (Same, but it predicts the signed rank of y.)	
Multiple regression: $\text{lm}(y \sim 1 + x_1 + x_2 + \dots)$	P: One-way ANOVA N: Kruskal-Wallis	aov(y ~ group) kruskal.test(y ~ group)	$\text{lm}(y \sim 1 + G_1 + G_2 + \dots + G_k)^a$ $\text{lm}(\text{rank}(y) \sim 1 + G_1 + G_2 + \dots + G_k)^a$	✓ for $N \geq 11$	An intercept for group 1 (plus a difference if group $\neq 1$) predicts y. - (Same, but it predicts the rank of y.)	
	P: One-way ANCOVA	aov(y ~ group + x)	$\text{lm}(y \sim 1 + G_1 + G_2 + \dots + G_k + x)^a$	✓	- (Same, but plus a slope on x.) <i>Note: this is discrete AND continuous. ANCOVAs are ANOVAs with a continuous x.</i>	
	P: Two-way ANOVA	aov(y ~ group * sex)	$\text{lm}(y \sim 1 + G_1 + G_2 + \dots + G_k + S_1 + S_2 + \dots + S_k + G_1^*S_1 + G_2^*S_2 + \dots + G_k^*S_k)$	✓	Interaction term: changing sex changes the y ~ group parameters. <i>Note: G_{ijk} is an indicator (0 or 1) for each non-intercept levels of the group variable. Similarly for S_{ijk} for sex. The first line (with G_1) is main effect of group, the second (with S_1) for sex and the third is the group * sex interaction. For two levels (e.g. male/female), line 2 would just be S_1 and line 3 would be S_1 multiplied with each G_i.</i>	[Coming]
	Counts ~ discrete x N: Chi-square test	chisq.test(groupXsex_table)	Equivalent log-linear model $\text{glm}(y \sim 1 + G_1 + G_2 + \dots + G_k + S_1 + S_2 + \dots + S_k + G_1^*S_1 + G_2^*S_2 + \dots + G_k^*S_k, \text{family} = \dots)^a$	✓	Interaction term: (Same as Two-way ANOVA.) <i>Note: Run glm using the following arguments: glm(model, family=poisson()) As linear-model, the Chi-square test is $\log(y) = \log(N) + \log(a) + \log(b) + \log(a\beta)$ where a and β are proportions. See more info in the accompanying notebook.</i>	Same as Two-way ANOVA
	N: Goodness of fit	chisq.test(y)	$\text{glm}(y \sim 1 + G_1 + G_2 + \dots + G_k, \text{family} = \dots)^a$	✓	(Same as One-way ANOVA and see Chi-Square note.)	1W-ANOVA

List of common parametric (P) non-parametric (N) tests and equivalent linear models. The notation $y \sim 1 + x$ is R shorthand for $y = 1 \cdot b + a \cdot x$ which most of us learned in school. Models in similar colors are highly similar, but really, notice how similar they all are across colors! For non-parametric models, the linear models are reasonable approximations for non-small sample sizes (see "Exact" column and click links to see simulations). Other less accurate approximations exist, e.g., Wilcoxon for the sign test and Goodness-of-fit for the binomial test. The signed rank function is `signed_rank = function(x) sign(x) * rank(abs(x))`. The variables G and S are "dummy coded" indicator variables (either 0 or 1) exploiting the fact that when $\Delta x = 1$ between categories the difference equals the slope. Subscripts (e.g., G_1 or y_1) indicate different columns in data. lm requires long-format data for all non-continuous models. All of this is exposed in greater detail and worked examples at <https://lindeloev.github.io/tests-as-linear>.

^a See the note to the two-way ANOVA for explanation of the notation.

^b Same model, but with one variance per group: `glm(value ~ 1 + G1, weights = varident(form = ~1|group), method="ML")`.



Jonas Kristoffer Lindeløv
<https://lindeloev.net>

<https://lindeloev.github.io/tests-as-linear/>

With GLM we can analyse
many different types of data
using many predictors
(quantitative & qualitative)

Unified, coherent framework for data analysis with many extensions:

- **GLMM** (mixed models): accomodate data structure & variation (space, time, phylogeny)

Unified, coherent framework for data analysis with many extensions:

- **GLMM** (mixed models): accomodate data structure & variation (space, time, phylogeny)
- **GAMM** (generalised additive models): non-linear relationships

Unified, coherent framework for data analysis with many extensions:

- **GLMM** (mixed models): accomodate data structure & variation (space, time, phylogeny)
- **GAMM** (generalised additive models): non-linear relationships
- **Model-based multivariate** statistics

Unified, coherent framework for data analysis with many extensions:

- **GLMM** (mixed models): accomodate data structure & variation (space, time, phylogeny)
- **GAMM** (generalised additive models): non-linear relationships
- **Model-based multivariate** statistics
- **Bayesian** modelling

The Generalised Linear Model (GLM) is a particularly reasonable vantage point on statistical analyses, as **many tests and procedures are special cases** of the GLM. The downside of that (and any other) vantage point is that **we first have to climb it**. There are the morass of unfamiliar terminology, the scree slopes of probability and the cliffs of distributions. **The vista, however, is magnificent.** From the GLM, t-test, ANOVA and regression neatly arrange themselves into regular patterns, and we can see the paths leading towards the horizon: to time series analyses, Bayesian statistics, spatial statistics and so forth.

[Dormann 2020](#)